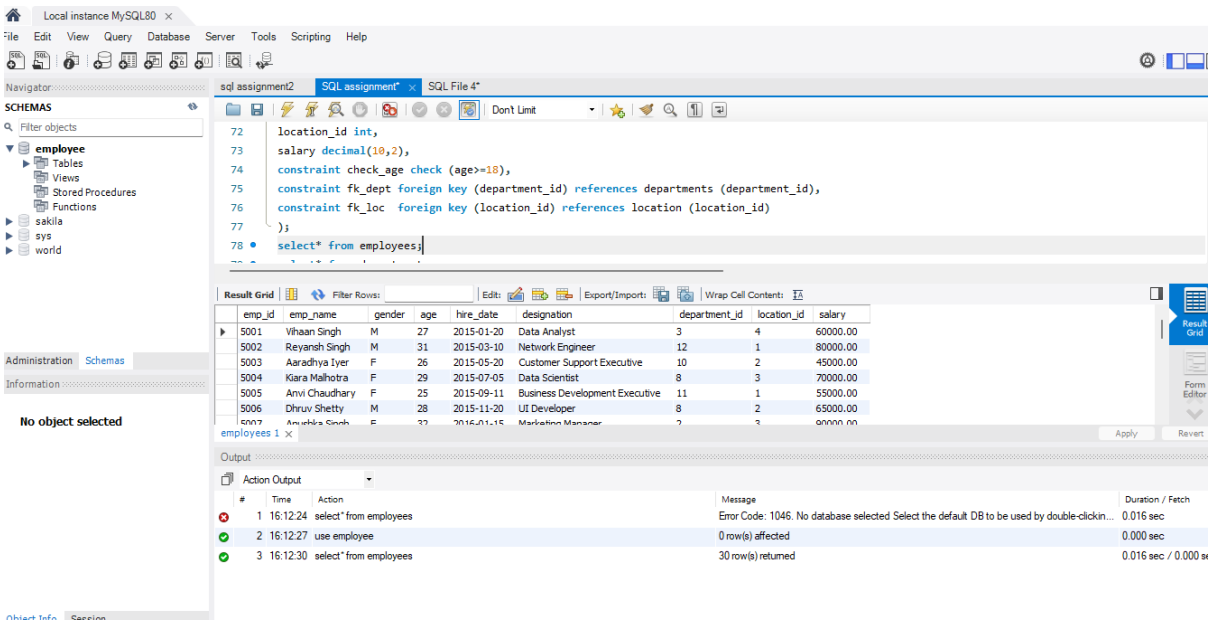
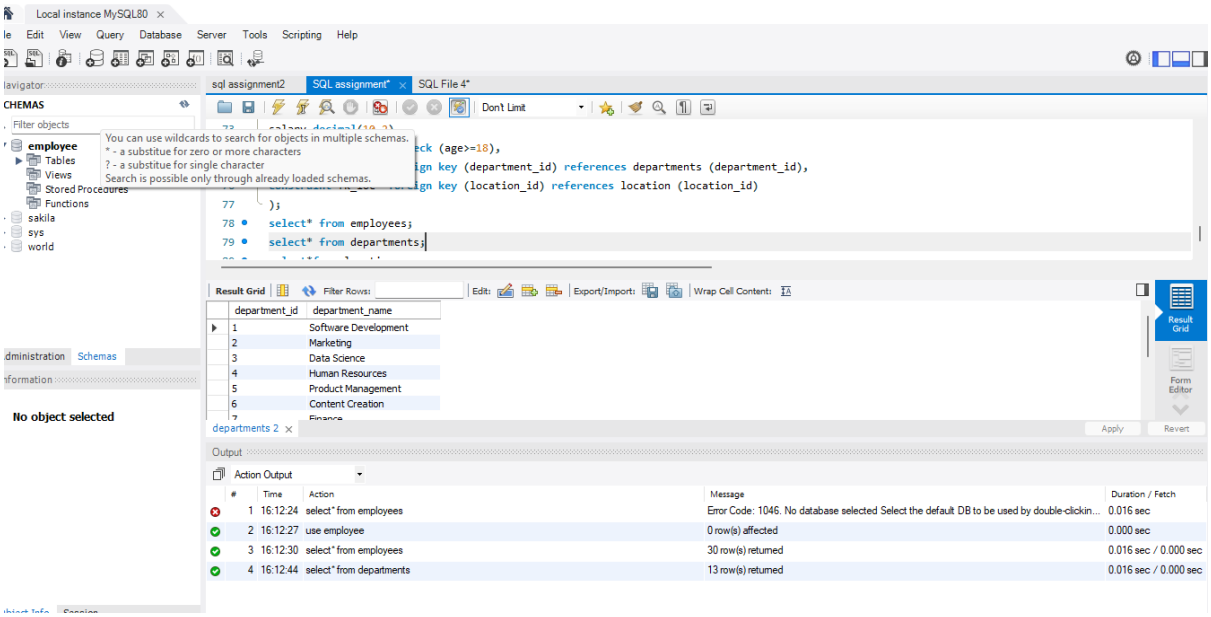
SQL ASSIGNMENT-2

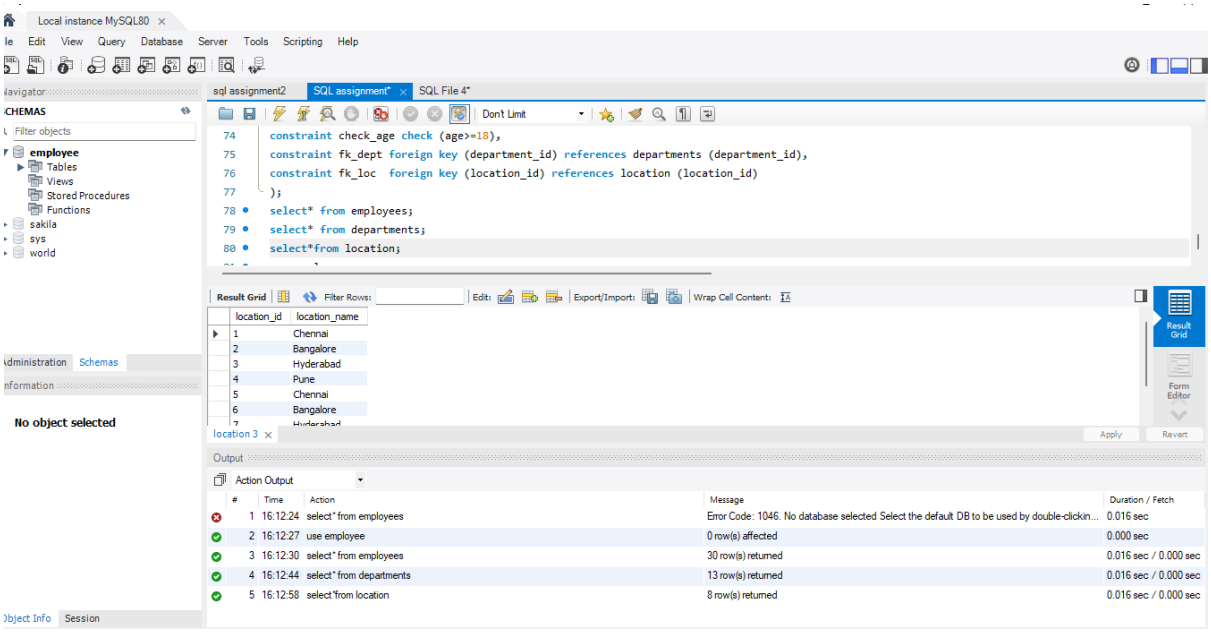
★ Continue from the previous assignment (database & table creation with constraints).

★ Use the provided DML insert codes to populate the tables from here: Employee Data

★ Execute the queries in this assignment after inserting the data.

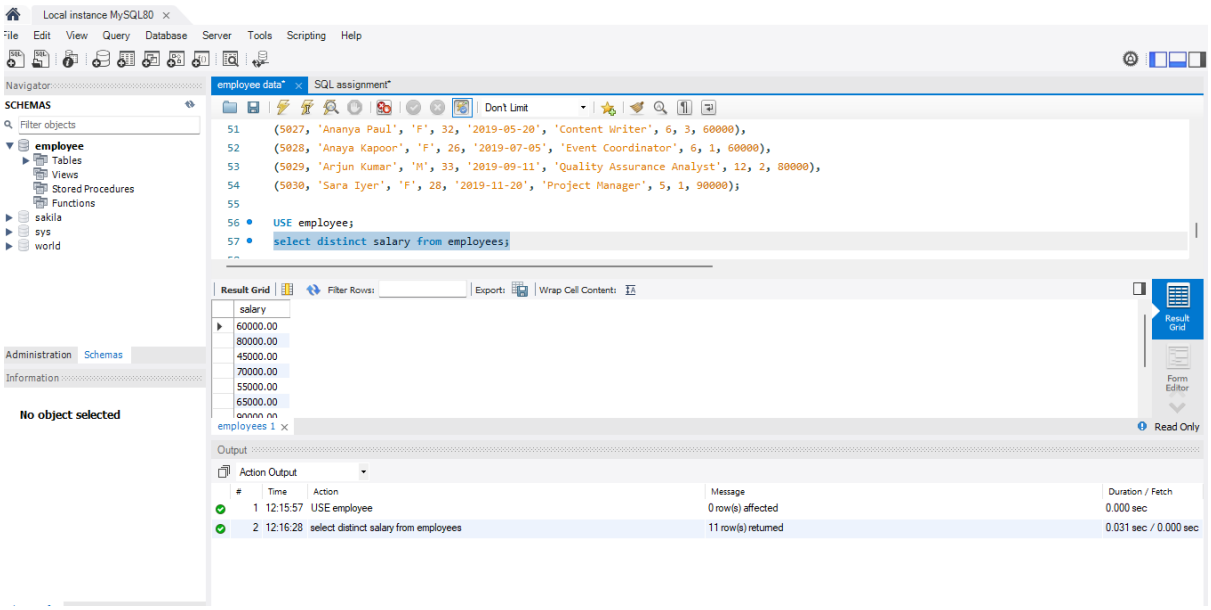






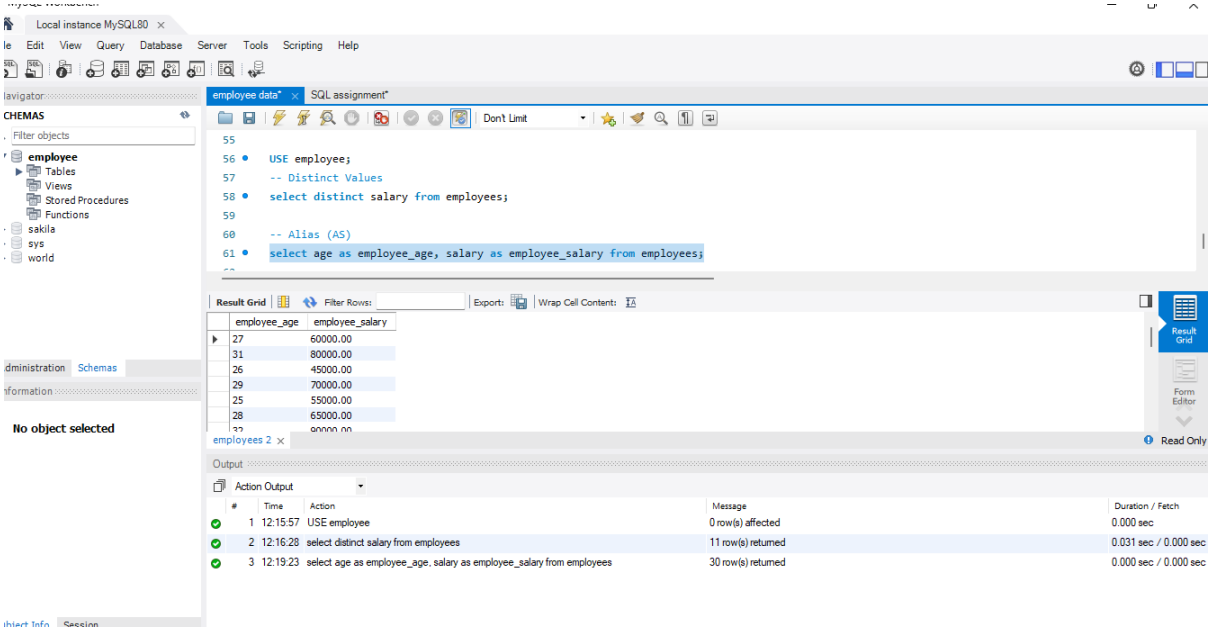
Distinct Values:

◉ a query to retrieve distinct salaries from the Employees table.



Alias (AS):

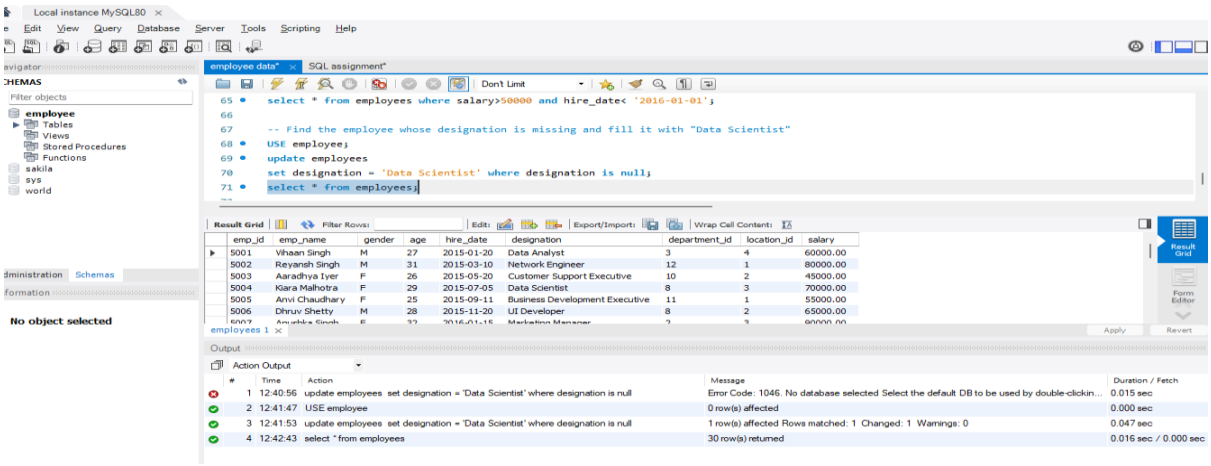
◉ Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

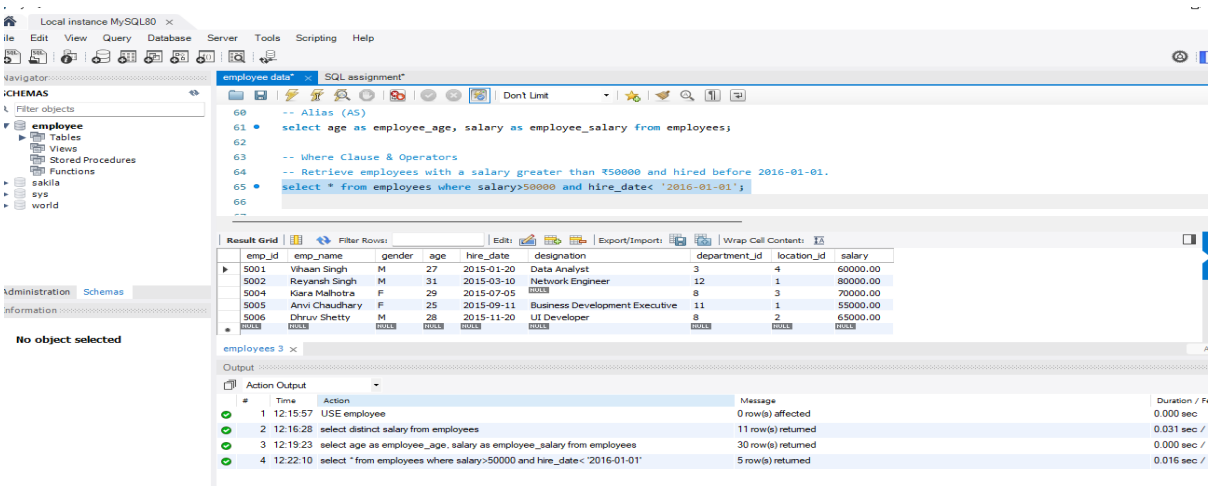


Where Clause & Operators:

◉ Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

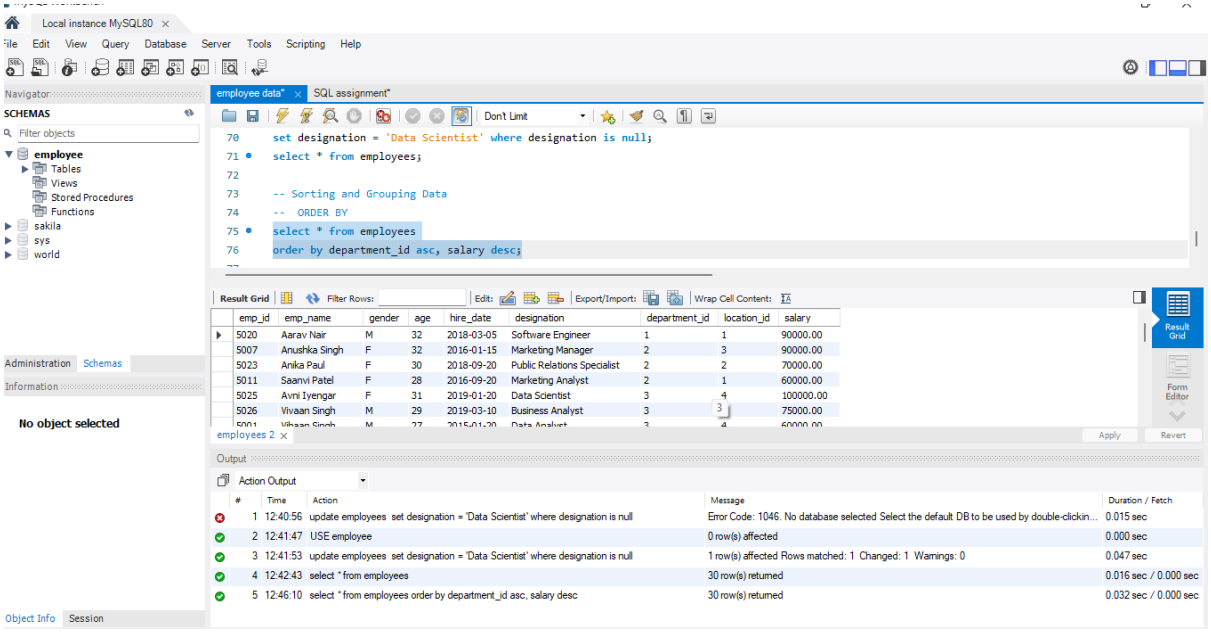
◉ Find the employee whose designation is missing and fill it with "Data Scientist".





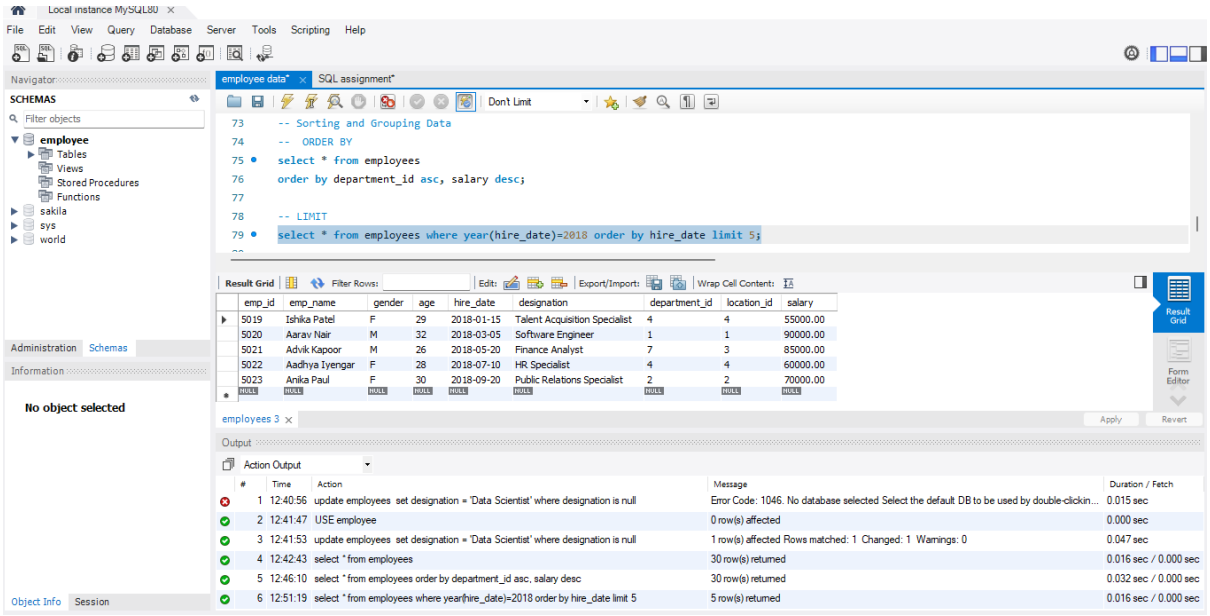
Sorting and Grouping Data:

1. ORDER BY: ◉ Find employees sorted by department ID in ascending order and salary in descending order.



2. LIMIT:

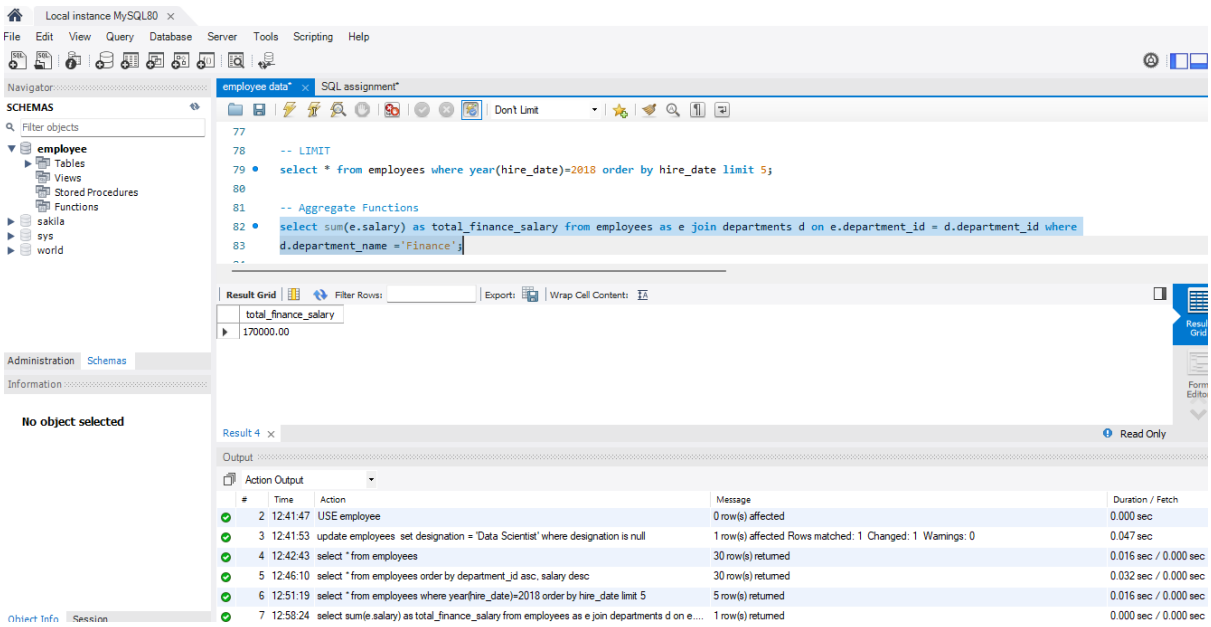
◉ Display the first 5 employees hired in the year 2018.

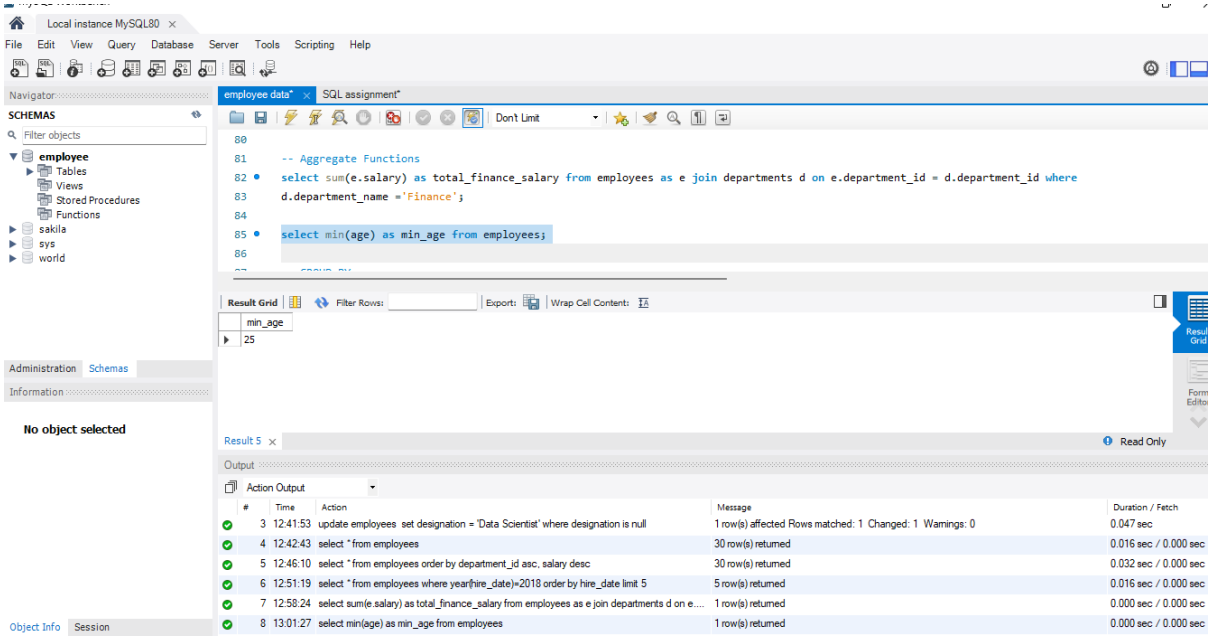


3. Aggregate Functions:

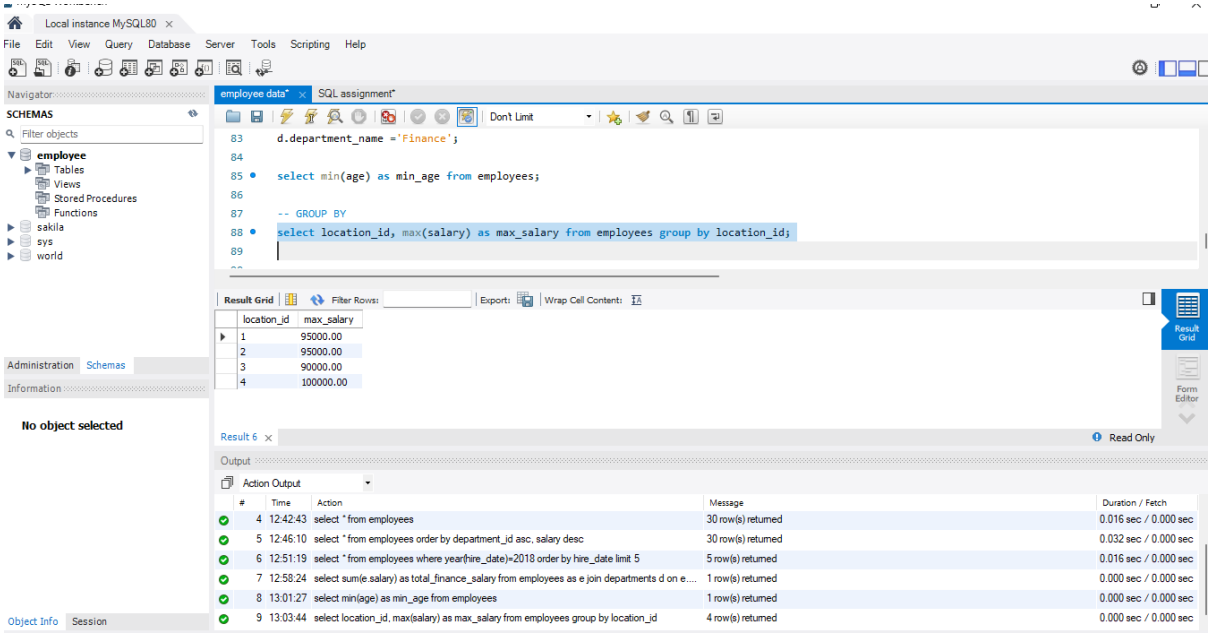
◉ Calculate the sum of all salaries in the Finance department.

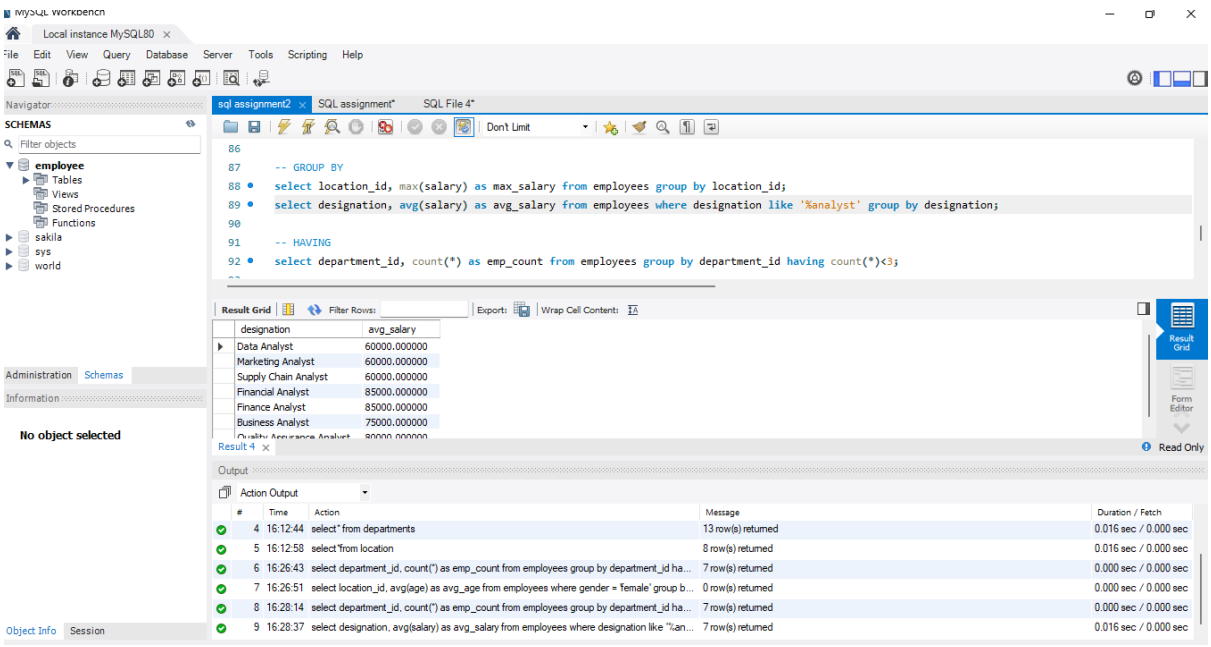
◉ Find the minimum age among all employees.





4. GROUP BY:
   ◉ List the maximum salary for each location.
   ◉ Calculate the average salary for each designation containing the word 'Analyst'.

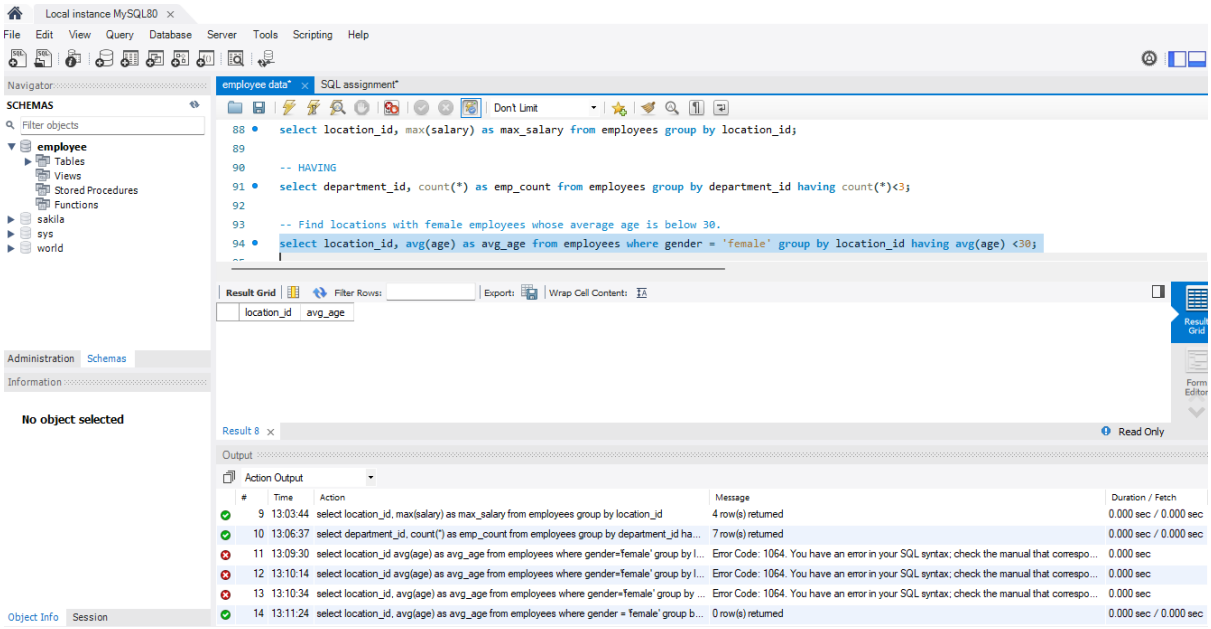

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No object selected
Object Info Session
employee data* SQL assignment*
Don't Limit
83 d.department_name ='Finance';
84
85 select min(age) as min_age from employees;
86
87 -- GROUP BY
88 select location_id, max(salary) as max_salary from employees group by location_id;
89
Result Grid Filter Rows: Export: Wrap Cell Content:
location_id max_salary
1 95000.00
2 95000.00
3 90000.00
4 100000.00
Result 6
Read Only
Output
Action Output
Time Action Message Duration / Fetch
4 12:42:43 select * from employees 30 row(s) returned 0.016 sec / 0.000 sec
5 12:46:10 select * from employees order by department_id asc, salary desc 30 row(s) returned 0.032 sec / 0.000 sec
6 12:51:19 select * from employees where year(hire_date)=2018 order by hire_date limit 5 5 row(s) returned 0.016 sec / 0.000 sec
7 12:58:24 select sum(e.salary) as total_finance_salary from employees as e join departments d on e.... 1 row(s) returned 0.000 sec / 0.000 sec
8 13:01:27 select min(age) as min_age from employees 1 row(s) returned 0.000 sec / 0.000 sec
9 13:03:44 select location_id, max(salary) as max_salary from employees group by location_id 4 row(s) returned 0.000 sec / 0.000 sec

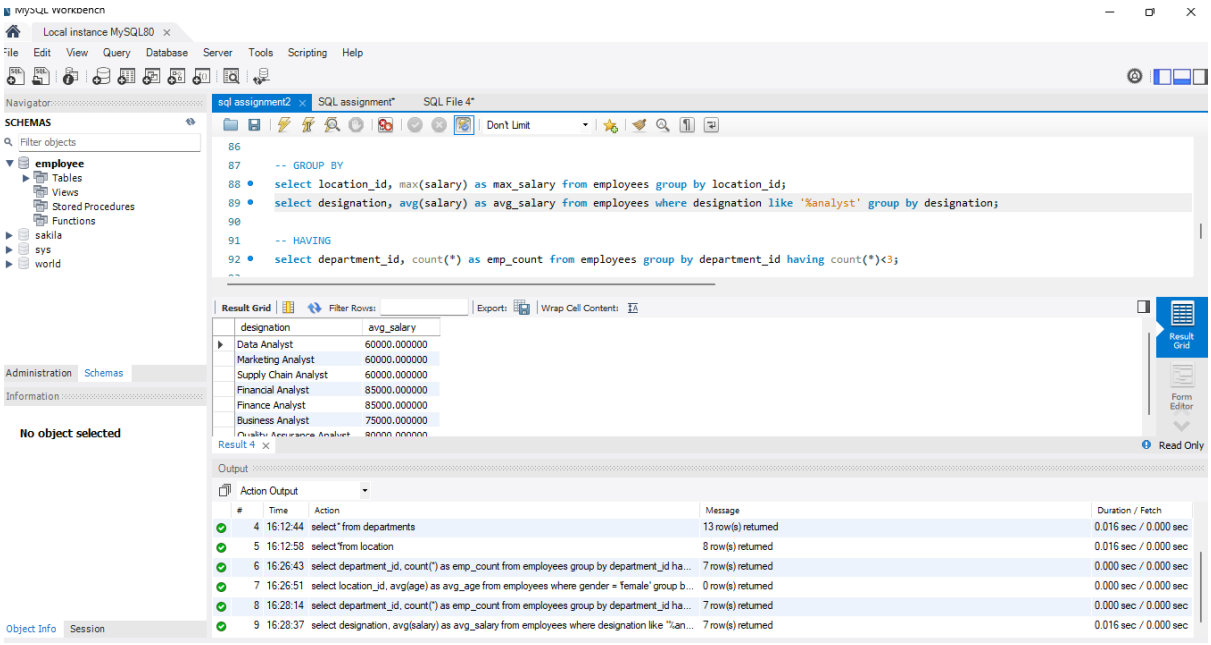

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sys
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No object selected
Object Info Session
sql assignment2 SQL assignment* SQL File 4*
Don't Limit
86
87 -- GROUP BY
88 select location_id, max(salary) as max_salary from employees group by location_id;
89 select designation, avg(salary) as avg_salary from employees where designation like '%analyst' group by designation;
90
91 -- HAVING
92 select department_id, count(*) as emp_count from employees group by department_id having count(*)<3;
Result Grid Filter Rows: Export: Wrap Cell Content:
designation avg_salary
Data Analyst 60000.000000
Marketing Analyst 60000.000000
Supply Chain Analyst 60000.000000
Financial Analyst 85000.000000
Finance Analyst 85000.000000
Business Analyst 75000.000000
Result 4
Read Only
Output
Action Output
Time Action Message Duration / Fetch
4 16:12:44 select * from departments 13 row(s) returned 0.016 sec / 0.000 sec
5 16:12:58 select *from location 8 row(s) returned 0.016 sec / 0.000 sec
6 16:26:43 select department_id, count(*) as emp_count from employees group by department_id ha... 7 row(s) returned 0.000 sec / 0.000 sec
7 16:26:51 select location_id, avg(age) as avg_age from employees where gender = 'female' group b... 0 row(s) returned 0.000 sec / 0.000 sec
8 16:28:14 select department_id, count(*) as emp_count from employees group by department_id ha... 7 row(s) returned 0.000 sec / 0.000 sec
9 16:28:37 select designation, avg(salary) as avg_salary from employees where designation like '%an... 7 row(s) returned 0.016 sec / 0.000 sec

HAVING:

◉ Find departments with less than 3 employees.

◉ Find locations with female employees whose average age is below 30.





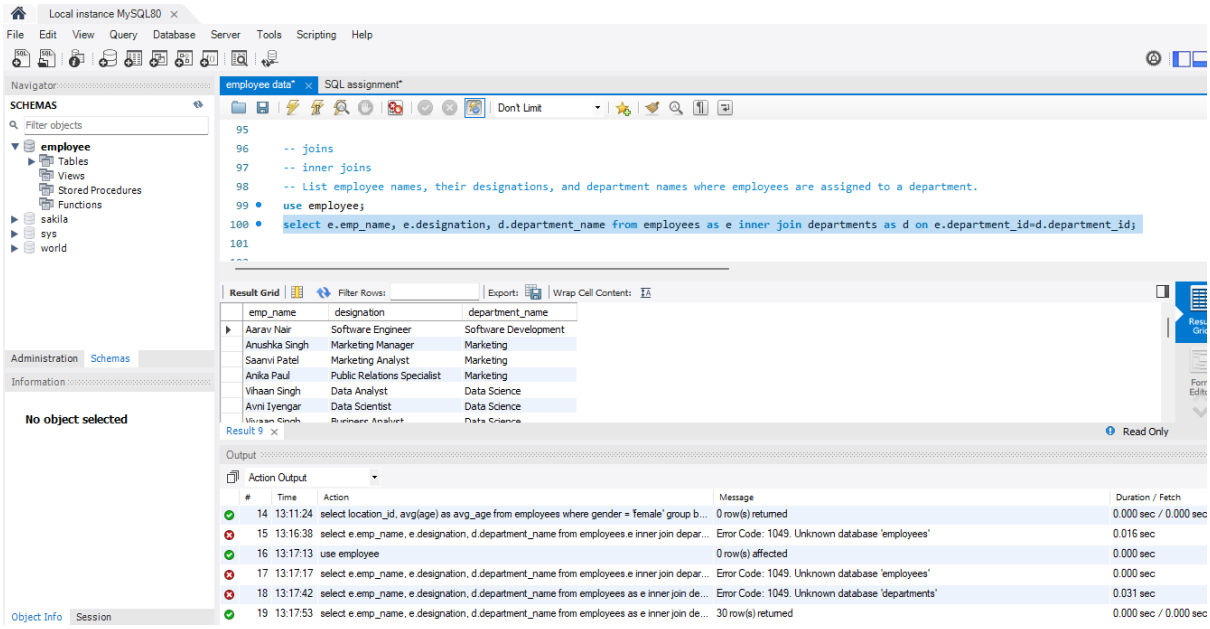
Joins: 1.

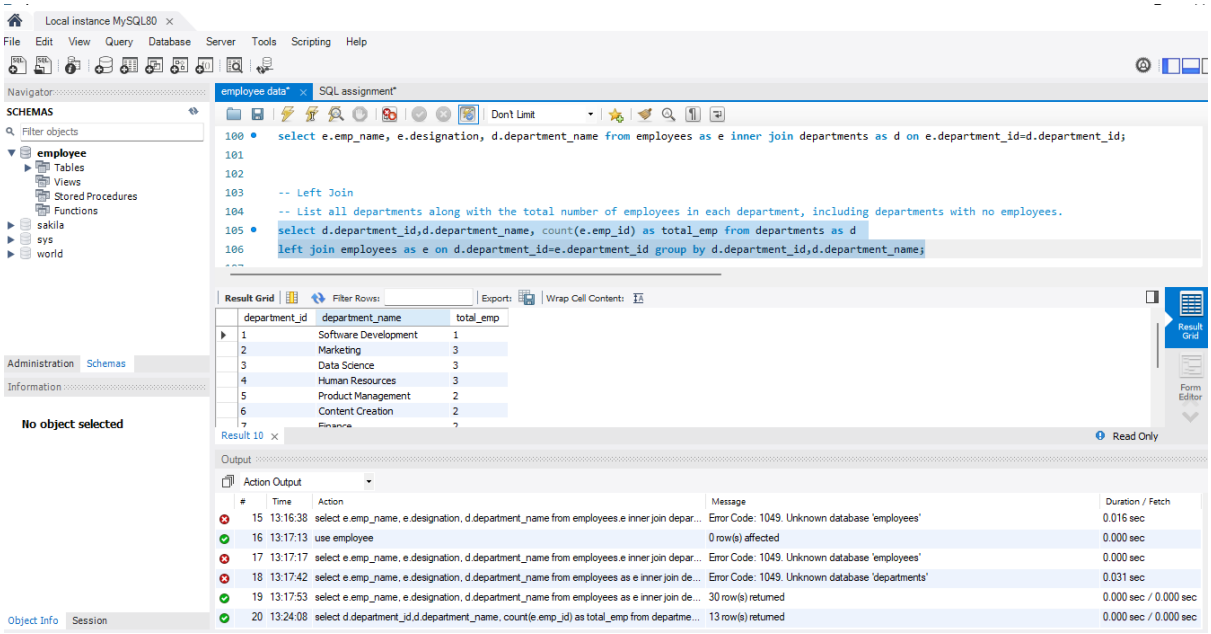
Inner Join:

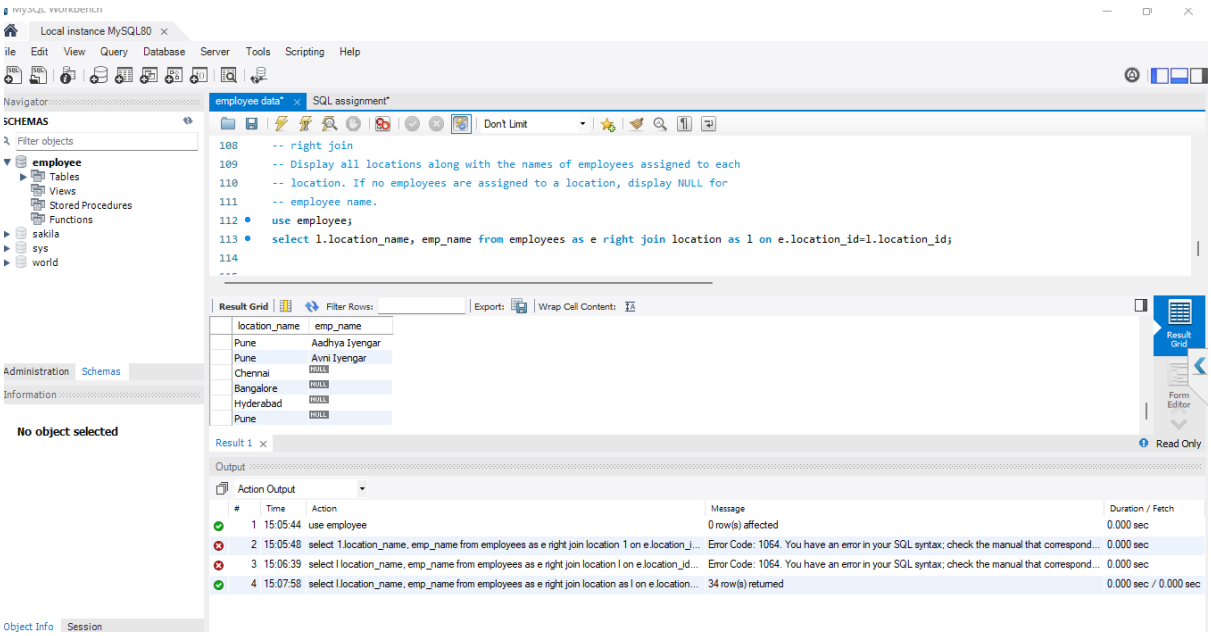
◉ List employee names, their designations, and department names where employees are assigned to a department.

2. Left Join: ◉ List all departments along with the total number of employees in each department, including departments with no employees.

3. Right Join: ◉ Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.




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No object selected
Object Info Session
employee data* SQL assignment*
Don't Limit
100 select e.emp_name, e.designation, d.department_name from employees as e inner join departments as d on e.department_id=d.department_id;
101
102
103 -- Left Join
104 -- List all departments along with the total number of employees in each department, including departments with no employees.
105 select d.department_id,d.department_name, count(e.emp_id) as total_emp from departments as d
106 left join employees as e on d.department_id=e.department_id group by d.department_id,d.department_name;
Result Grid Filter Rows: Export: Wrap Cell Content:
department_id department_name total_emp
1 Software Development 1
2 Marketing 3
3 Data Science 3
4 Human Resources 3
5 Product Management 2
6 Content Creation 2
Result 10
Read Only
Output
Action Output
Time Action Message Duration / Fetch
15 13:16:38 select e.emp_name, e.designation, d.department_name from employees.e inner join depar... Error Code: 1049. Unknown database 'employees' 0.016 sec
16 13:17:13 use employee 0 row(s) affected 0.000 sec
17 13:17:17 select e.emp_name, e.designation, d.department_name from employees.e inner join depar... Error Code: 1049. Unknown database 'employees' 0.000 sec
18 13:17:42 select e.emp_name, e.designation, d.department_name from employees as e inner join de... Error Code: 1049. Unknown database 'departments' 0.031 sec
19 13:17:53 select e.emp_name, e.designation, d.department_name from employees as e inner join de... 30 row(s) returned 0.000 sec / 0.000 sec
20 13:24:08 select d.department_id,d.department_name, count(e.emp_id) as total_emp from departme... 13 row(s) returned 0.000 sec / 0.000 sec


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Information
No object selected
Object Info Session
employee data* SQL assignment*
Don't Limit
108 -- right join
109 -- Display all locations along with the names of employees assigned to each
110 -- location. If no employees are assigned to a location, display NULL for
111 -- employee name.
112 use employee;
113 select l.location_name, emp_name from employees as e right join location as l on e.location_id=l.location_id;
114
Result Grid Filter Rows: Export: Wrap Cell Content:
location_name emp_name
Pune Aadhya Iyengar
Pune Avni Iyengar
Chennai NULL
Bangalore NULL
Hyderabad NULL
Pune NULL
Result 1
Read Only
Output
Action Output
Time Action Message Duration / Fetch
1 15:05:44 use employee 0 row(s) affected 0.000 sec
2 15:05:48 select l.location_name, emp_name from employees as e right join location l on e.location_i... Error Code: 1064. You have an error in your SQL syntax; check the manual that correspond... 0.000 sec
3 15:06:39 select l location_name, emp_name from employees as e right join location l on e.location_id... Error Code: 1064. You have an error in your SQL syntax; check the manual that correspond... 0.000 sec
4 15:07:58 select l.location_name, emp_name from employees as e right join location as l on e.location... 34 row(s) returned 0.000 sec / 0.000 sec